

## **Course Name: Object Oriented programming language**

**Course Code:**      **Internal Marks: 25**      **External Marks: 75**      **Credit: 4**

### **UNIT-I**

**Introduction:-** Introducing Object – Oriented Approach, Relating to other paradigms {Functional, Data decomposition}. Basic terms and ideas Abstraction, Encapsulation, Inheritance, Polymorphism, Review of C, Difference between C and C++ - cin, cout, new, delete, operators.

### **UNIT-II**

**Class and Object:-** Encapsulation, information hiding, abstract data types, Object & classes, attributes, methods, C++ class declaration, State identity and behaviour of an object, Constructors and destructors, instantiation of objects, Default parameter value, object types, C++ garbage collection, dynamic memory allocation, Metaclass / abstract classes.

### **UNIT-III**

**Inheritance and Polymorphism:-** Inheritance, Class hierarchy, derivation – public, private & protected, Aggregation, composition vs classification hierarchies, Polymorphism, Categorization of polymorphism techniques, Method polymorphism, Polymorphism by parameter, Operator overloading, Parameteric Polymorphism.

### **UNIT-IV**

**Generic Function:-** Template function, function name overloading, overriding inheritance methods, Run time polymorphism, Multiple Inheritance.

### **UNIT-V**

**Files and Exception Handling:-** Streams and files, Namespaces, Exception handling, Generic Classes.